KAIZEN MATHS

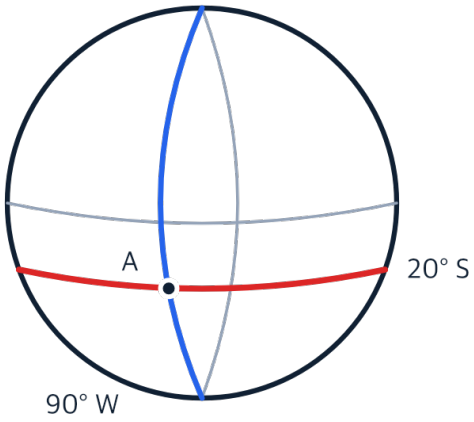
Lesson starter

Earth Geometry

Latitude and Longitude

Name _____

SA1



Schematic globe; points lie on the shown latitude or meridian

Point A is shown on the parallel 20° S and the longitude 90° W. Write the coordinates of A.

GEO-F170-L1 · SA

Show your working

KAIZEN MATHS

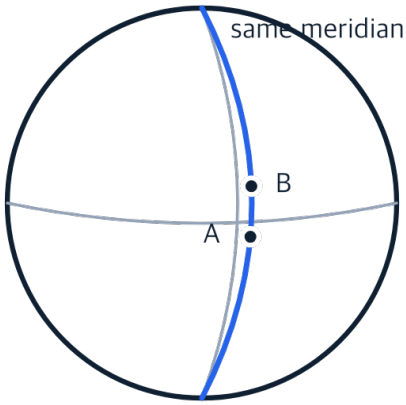
Lesson starter

Earth Geometry

Latitude and Longitude

Name _____

SB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the same meridian. A is at 10° S and B is at 5° N. Taking the Earth radius as 6400 km, find the distance from A to B along the meridian.

GEO-F170-L1 · SB

Show your working

KAIZEN MATHS

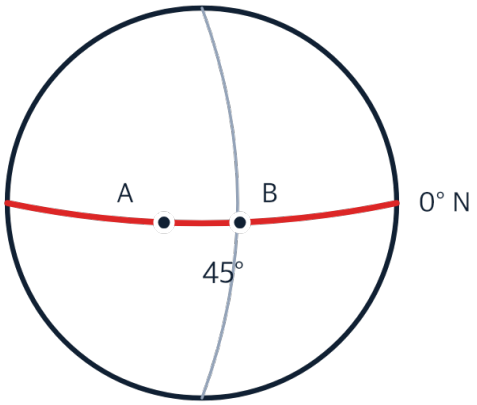
Lesson starter

Earth Geometry

Latitude and Longitude

Name _____

SC1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the equator and their longitudes differ by 45°. Taking the Earth radius as 6400 km, find the shorter distance from A to B along the equator.

GEO-F170-L1 · SC

Show your working

KAIZEN MATHS

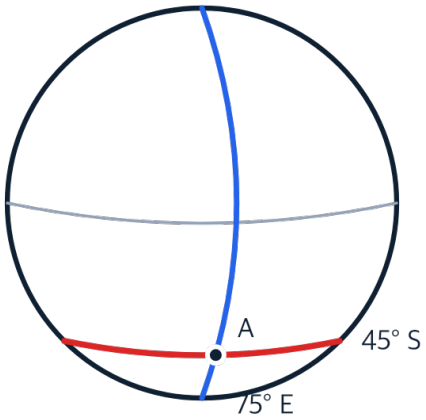
Lesson starter

Earth Geometry

Latitude and Longitude

Name _____

SD1



Schematic globe; points lie on the shown latitude or meridian

Point A is shown on the parallel 45° S and the longitude 75° E. Write the coordinates of A.

GEO-F170-L1 · SD

Show your working

KAIZEN MATHS

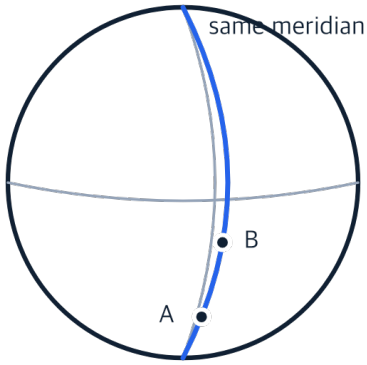
Exit ticket

Earth Geometry

Latitude and Longitude

Name _____

EA1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the same meridian. A is at 50° S and B is at 20° S. Taking the Earth radius as 6400 km, find the distance from A to B along the meridian.

GEO-F170-L1 · EA

Show your working

KAIZEN MATHS

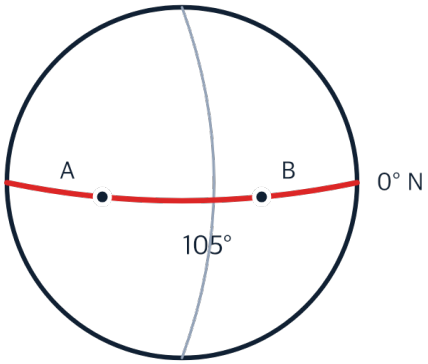
Exit ticket

Earth Geometry

Latitude and Longitude

Name _____

EB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the equator and their longitudes differ by 105°. Taking the Earth radius as 6400 km, find the shorter distance from A to B along the equator.

GEO-F170-L1 · EB

Show your working

KAIZEN MATHS

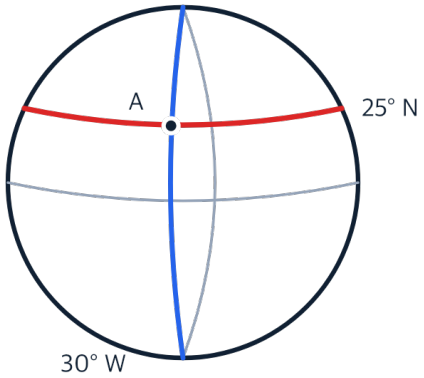
Exit ticket

Earth Geometry

Latitude and Longitude

Name _____

EC1



Schematic globe; points lie on the shown latitude or meridian

Point A is shown on the parallel 25° N and the longitude 30° W. Write the coordinates of A.

GEO-F170-L1 · EC

Show your working

KAIZEN MATHS

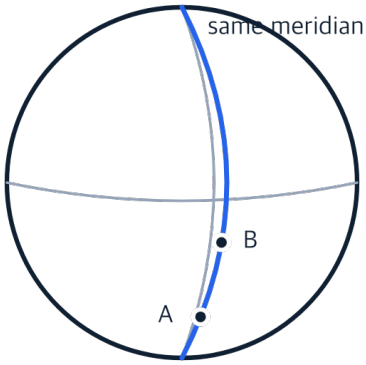
Exit ticket

Earth Geometry

Latitude and Longitude

Name _____

EA1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the same meridian. A is at 50° S and B is at 20° S. Taking the Earth radius as 6400 km, find the distance from A to B along the meridian.

GEO-F170-L1 · EA

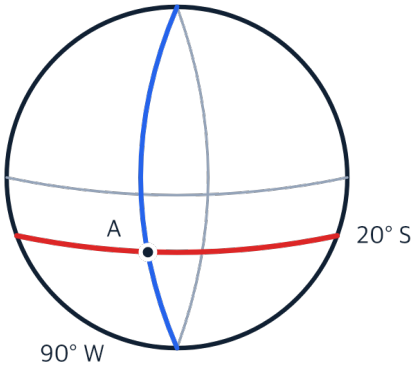
Show your working

Teacher answers & working

Earth Geometry · Latitude and Longitude

GEO-F170-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Schematic globe; points lie on the shown latitude or meridian

Point A is shown on the parallel 20° S and the longitude 90° W. Write the coordinates of A.

Answer:

20° S, 90° W

Working:

Step 1: Latitude is written first.

Step 2: The latitude is 20° S.

Step 3: Longitude is written second.

Step 4: The longitude is 90° W.

Coordinates of A = (20° S, 90° W).

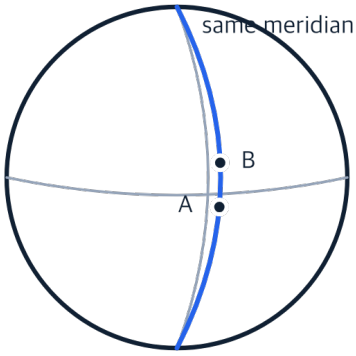
Final answer: 20° S, 90° W

Teacher answers & working

Earth Geometry · Latitude and Longitude

GEO-F170-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the same meridian. A is at 10° S and B is at 5° N. Taking the Earth radius as 6400 km, find the distance from A to B along the meridian.

Answer:

1,676 km

Working:

Step 1: The points are on the same meridian, so the path is part of a great circle.

Step 2: Find the latitude difference.

$|5 - -10| = 15°$

Step 3: Use arc length: distance = (angle ÷ 360) × 2πR.

distance = (15 ÷ 360) × 2π × 6400

distance ≈ 1675.5 km

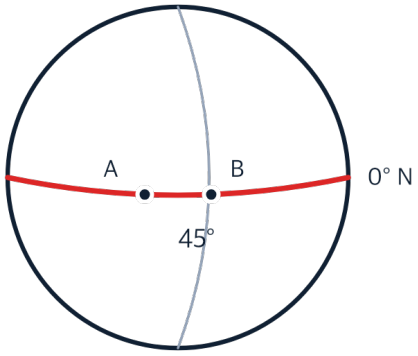
Final answer: 1,676 km

Teacher answers & working

Earth Geometry · Latitude and Longitude

GEO-F170-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the equator and their longitudes differ by 45°. Taking the Earth radius as 6400 km, find the shorter distance from A to B along the equator.

Answer:

5,027 km

Working:

Step 1: The equator is a great circle, so use radius R = 6400 km.

Step 2: The longitude difference is 45°.

Step 3: Use arc length: distance = (angle ÷ 360) × 2πR.

distance = (45 ÷ 360) × 2π × 6400

distance ≈ 5026.5 km

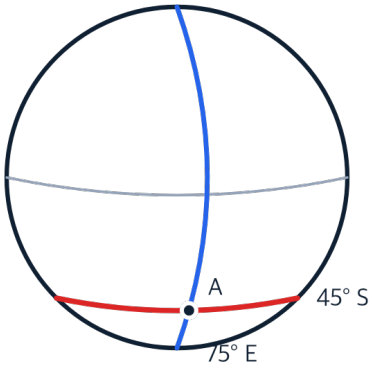
Final answer: 5,027 km

Teacher answers & working

Earth Geometry · Latitude and Longitude

GEO-F170-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Schematic globe; points lie on the shown latitude or meridian

Point A is shown on the parallel 45° S and the longitude 75° E. Write the coordinates of A.

Answer:

45° S, 75° E

Working:

Step 1: Latitude is written first.

Step 2: The latitude is 45° S.

Step 3: Longitude is written second.

Step 4: The longitude is 75° E.

Coordinates of A = (45° S, 75° E).

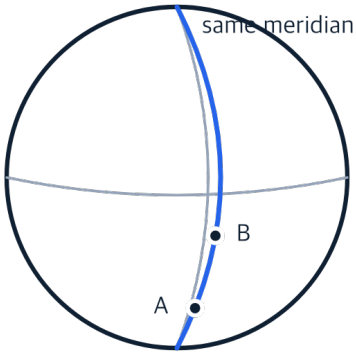
Final answer: 45° S, 75° E

Teacher answers & working

Earth Geometry · Latitude and Longitude

GEO-F170-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the same meridian. A is at 50° S and B is at 20° S. Taking the Earth radius as 6400 km, find the distance from A to B along the meridian.

Answer:

3,351 km

Working:

Step 1: The points are on the same meridian, so the path is part of a great circle.

Step 2: Find the latitude difference.

$|-20 - -50| = 30°$

Step 3: Use arc length: distance = (angle ÷ 360) × 2πR.

distance = (30 ÷ 360) × 2π × 6400

distance ≈ 3351.0 km

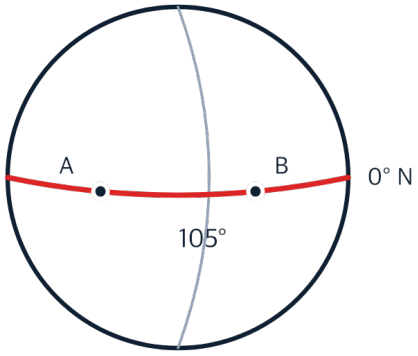
Final answer: 3,351 km

Teacher answers & working

Earth Geometry · Latitude and Longitude

GEO-F170-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on the equator and their longitudes differ by 105°. Taking the Earth radius as 6400 km, find the shorter distance from A to B along the equator.

Answer:

11,729 km

Working:

Step 1: The equator is a great circle, so use radius R = 6400 km.

Step 2: The longitude difference is 105°.

Step 3: Use arc length: distance = (angle ÷ 360) × 2πR.

distance = (105 ÷ 360) × 2π × 6400

distance ≈ 11728.6 km

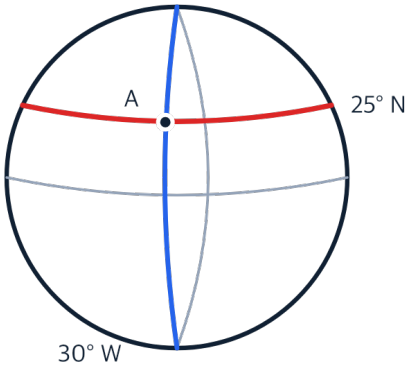
Final answer: 11,729 km

Teacher answers & working

Earth Geometry · Latitude and Longitude

GEO-F170-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Schematic globe; points lie on the shown latitude or meridian

Point A is shown on the parallel 25° N and the longitude 30° W. Write the coordinates of A.

Answer:

25° N, 30° W

Working:

Step 1: Latitude is written first.

Step 2: The latitude is 25° N.

Step 3: Longitude is written second.

Step 4: The longitude is 30° W.

Coordinates of A = (25° N, 30° W).

Final answer: 25° N, 30° W